Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Saarang Gaggar	Email	Saarang.gaggar@tno.nl
Institute Name	Netherlands Organization for Applied Scientific Research (TNO)			
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2. Technology Overview

Title of Proposed Technology	**V2X Collaborative Perception for Increased Road Safety**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication (√), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Vehicle Intelligence Technologies	Automobile / Rail Vehicles	Safety Improvement Technologies
Concept of Proposed Technology	Automated and connected driving faces significant challenges in complex environments such as intersections, dense urban areas, and scenarios involving vulnerable road users (VRUs). In such cases, the vehicle's onboard sensors may have limited visibility caused by occlusions from buildings, large vehicles, or roadside objects. This constraint reduces situational awareness and limits the safety envelope of automated driving systems. Collaborative perception (CP), enabled through Vehicle-to-Everything (V2X) communication, offers a solution by allowing vehicles and infrastructure to share their perceived environment. Information from roadside LiDAR sensors or other connected vehicles is transmitted to the ego (automated) vehicle in real time, expanding its effective field of view beyond the sensing range of onboard			

sensors alone. This enhances situational awareness, thereby improving decision-making, and safety performance.

TNO has developed a collaborative perception framework over multiple research programs, including the EcoMobility EU project. This framework integrates multiple perception sources into a shared world model, enabling automated vehicles to account for objects detected by both the ego, as well as those shared by other sources.

In April 2025, TNO successfully demonstrated this technology at the Drive Forward conference in the Netherlands, showing how a vehicle, despite having an occluded view of an intersection, could respond safely to a pedestrian detected around the intersection and communicated by another roadside (parked) vehicle.

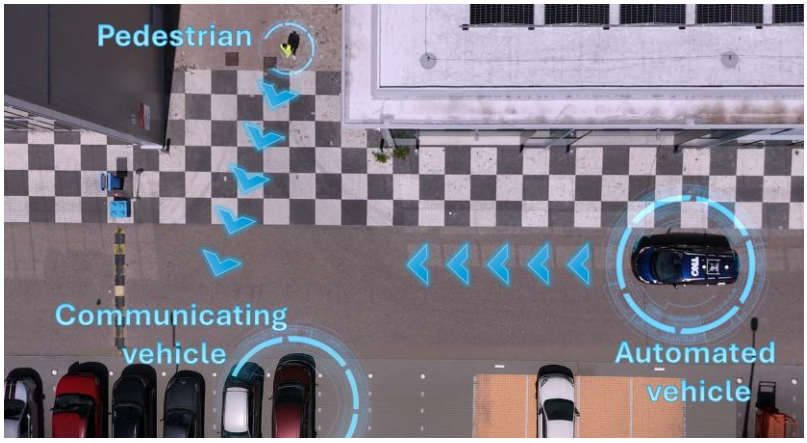

Image of the collaborative perception demonstration by TNO, during the Drive Forward conference in the Netherlands, in April 2025. The blue vehicle, being driven in a fully autonomous mode, aims to turn right next to the building. Even though it cannot see the pedestrian in its path through its own sensors, the automated vehicle comes to a halt due to a communicated message from the "Communicating Vehice" parked on the side of the road.

Building on this foundation, the proposed joint project with a Korean SME aims to mature the collaborative perception system by integrating mature vehicle and infrastructure LiDAR perception solutions, extending message specifications, improving synchronization and latency performance, and demonstrating real-world benefits across different deployment environments.

Core components to be developed include:

- Integration of vehicle-side and infrastructure-side LiDAR perception systems.
- Extension of collaborative perception message definitions.
- Development of a scalable shared world-model capable of fusing object information received through multiple sources.
- Deployment of the system on both road side units (RSUs) and (automated) vehicles.
- Validation of performance through demonstrations in Korea and the Netherlands.

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Rationale for Joint R&D Project	A joint R&D project is necessary to combine complementary strengths and ensure that collaborative perception matures towards validated smart-mobility technology. **Technical Need:** A practical, validated cooperative perception system needs to: • Fuse vehicle LiDAR detections and infrastructure LiDAR detections in real time. • Transmit object-level information reliably via V2X. • Ensure synchronization, latency control, and security. • Meet interoperability requirements across different ecosystems. **Complementarity:** • **TNO** brings expertise in (shared) world modelling, V2X communication and message design, ITS integration, definition of AD interfaces, system architecture design, and access to controlled proving grounds and dedicated test facilities. • **Korean SME** brings production-grade LiDAR perception technology, infrastructure based perception, and experience with real-world smart-city deployments.
Key Words	1. Collaborative (Collective) Perception 2. LiDAR Perception 3. VRU Protection 4. Shared World Model 5. C-ITS / CCAM Pilots

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	Develop a scalable shared world-model framework capable of fusing vehicle-based and infrastructure-based LiDAR perception outputs. • Define and standardize collaborative perception message formats for real-time V2X communication. • Integrate the SME's vehicle and infrastructure perception solutions with TNO's collaborative perception framework. • Improve performance on communication latency, sensor noise etc. • Conduct integrated demonstrations in Korea and the Netherlands to validate system performance, safety benefits, and deployment feasibility.			
R&D Target Specifications	• **Increase FoV** for intersection and heavily occluded scenarios by 20%. • **Communication latency:** Achieve end-to-end (perception → V2X communication → fusion) latency of ≤0.5s for safety-critical objects. • **World-model consistency:** Maintain object-track alignment mismatch of ≤0.5 m across fused sources. • **System availability:** ≥ 99% during field demonstrations.			

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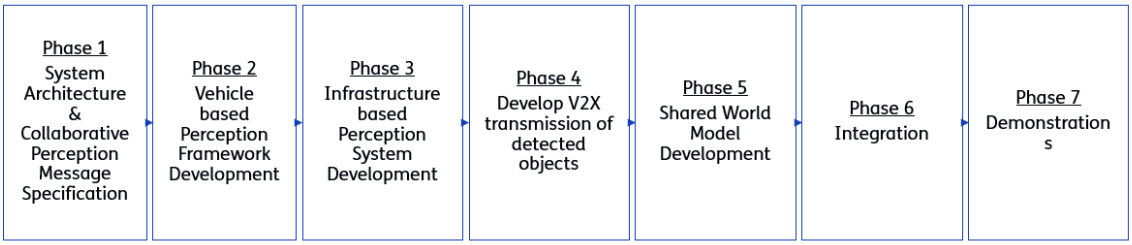
R&D Plan

TNO Proposal: V2X Collaborative Perception for Increased Road Safety



Phase 1: System Architecture & Collaborative Perception Message Specification (Months 0-6; Lead: TNO)

- Define system requirements and target scenarios.
- Specify overall system architecture, interfaces, and collaborative perception message structure.
- Align perception outputs from the SME with TNO's fusion framework.

Milestone M1: Complete system architecture, interface descriptions, and collaborative perception message definitions.

Phase 2: Vehicle-based Perception Framework Development (Months 6–12; Lead: SME)

- Enhance vehicle-based LiDAR perception to generate track-level object data.

Milestone M2: Prototype vehicle-side LiDAR based perception module.

Phase 3: Infrastructure-based Perception System Development (Months 6–12; Lead: SME)

- Enhance infrastructure LiDAR perception to generate track-level object data.

Milestone M3: Prototype RSU-side perception and V2X communication.

Phase 4: Develop V2X transmission of detected objects (Months 6-9; Lead: TNO)

- Implement real-time V2X transmission of detected objects.

Milestone M4: V2X transmission module (hardware and software)

Phase 5: Shared World-Model Development (Months 6–14; Lead: TNO)

- Develop and implement the shared world model framework.
- Ensure compatibility with both vehicle and RSU perception streams.

Milestone M5: Operational shared world-model framework.

Phase 6: Integration (Months 14–18; Leads: Korea: SME; Netherlands: TNO)

- Integrate modules on SME's vehicle platform and TNO's Carlab test setup.
- Deploy collaborative perception on roadside units at SME's site, and at TNO.
- Conduct iterative testing for interoperability, robustness, and real-time performance.

Milestone M6: Integrated system validated in controlled environments.

Phase 7: Demonstrations (Months 18–24; Leads: Korea: SME; Netherlands: TNO)

- Execute demonstration scenarios featuring occluded VRU crossings, intersection approaches, and mixed-traffic conditions.
- Collect performance data; TNO analyses results and prepares evaluation reports.

Milestone M7: Demonstration report with KPIs and deployment recommendations.

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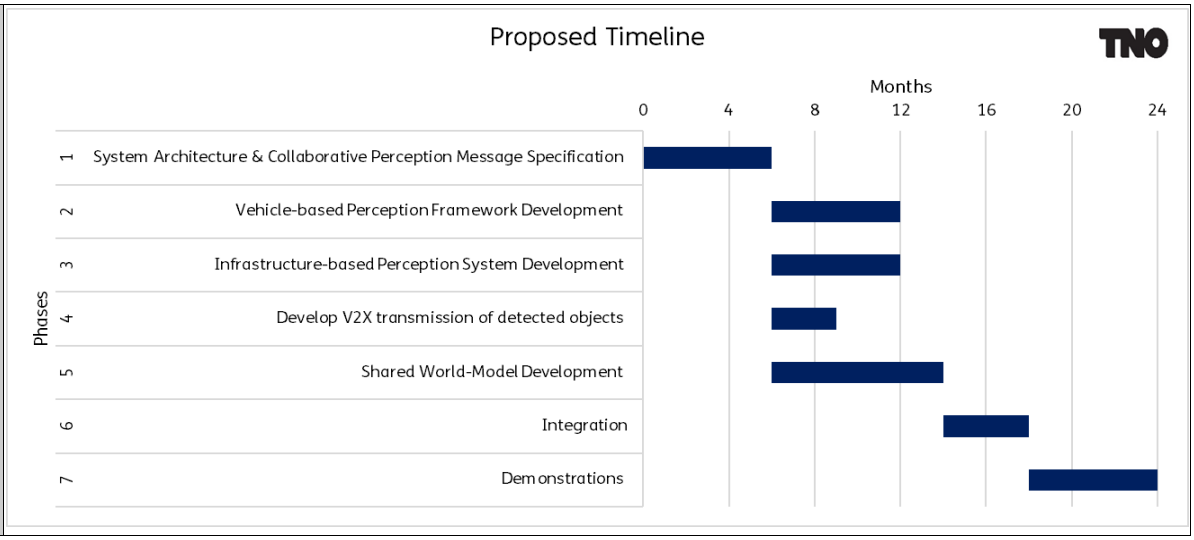

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead system architecture, CP message specification, and shared world-model development. • Develop and maintain V2X message handling. • Support integration on RSUs and vehicle-side in the Netherlands. • Provide test facilities (Carlab, MARQ) for integration testing and validation. • Analyze Korean demonstration results and generate final technical documentation.
Expected R&R of Korean SME Partner	• Provide vehicle and infrastructure LiDAR perception systems. • Develop interfaces for transmitting and receiving collaborative perception data. • Host and manage the Korean demonstration site, including infrastructure and vehicle deployment. • Conduct data collection, system testing, and initial validation. • Collaborate on performance tuning and scenario refinement.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint project is expected to deliver:
	• A TRL-6 collaborative perception platform validated in two countries, demonstrating real-world safety and situational awareness benefits.
	• Improved VRU protection through early detection in occluded intersections, enhancing safety for pedestrians and cyclists.
	• Strong commercialization potential for
	  ○ Smart-city deployments using RSU-based LiDAR
	  ○ OEM and Tier-1 adoption of CP-enhanced ADAS/AD stacks

6. References

(1) D2.2 System Design for Enhanced Perception and Localisation – EU Ecomobility Project; Hennecke Marcus; Projec Report 2024.